

## Experiment 1:

### PERIPHERALS OF A COMPUTER (INPUT/OUTPUT DEVICES)

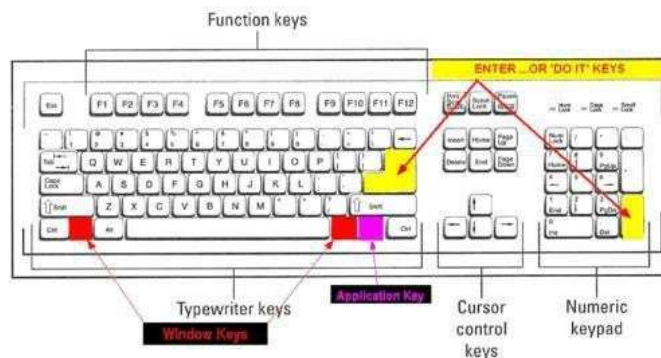
A computer peripheral refers to any external device that facilitates the interaction between a user and a computer system by either providing input or output functionalities. Input peripherals, like keyboards and mice, enable users to input data into the computer, while output peripherals, such as monitors and printers, display or produce information generated by the computer. Additionally, certain peripherals, like external hard drives, offer both input and output capabilities, enhancing the storage and transfer of data between the computer and external sources.

#### INPUT DEVICES:

These devices convert data into a binary format that the computer can understand and process. They transmit this binary data to the computer for further processing and also receive data from external sources. Common examples of such input devices include keyboards, mice, joysticks, trackballs, and scanners.

#### Keyboard:

The keyboard stands out as one of the most prevalent input devices, allowing users to manually input data through typing. Resembling a typewriter keyboard, it features additional specialized keys, such as function keys and control keys. These keys can be customized by users to suit their specific requirements, offering a personalized and adaptable interface for interacting with the computer system.



#### Mouse:

The mouse functions primarily as a pointing device, serving as an input tool for transmitting the coordinates of the screen's cursor position. Unlike keyboards that input characters, the mouse interacts with graphical user interface (GUI) software by relaying precise location data. This enables users to navigate interfaces, select items, and execute commands efficiently, particularly within GUI-based

software applications.



### **Joystick And Tracker ball:**

Joysticks and trackballs also function as pointing devices, akin to the mouse, serving a similar purpose in facilitating user interaction with graphical interfaces. These devices allow users to manipulate on-screen elements by providing directional control or precise cursor movement, offering alternative input options for navigating and interacting with software applications.



### **Scanner:**

Scanners capture comprehensive image data directly from their source, typically a page, and store it in a graphic format suitable for display on screens. This process enables the reproduction of detailed visual content, facilitating the digitalization of physical documents or images for viewing and manipulation on electronic devices.



**Touch Pads:** Touch pads are tactile devices positioned on desktop surfaces, designed to detect and react to applied pressure.



### **Bar Code Readers:**

Barcoded data finds common application in labeling goods and numbering books. A Barcode Reader is instrumental in this process, scanning the barcode image and converting it into an alphanumeric value. This value is then transmitted to the connected computer, facilitating efficient data entry and management.



### **Optical Mark Reader (OMR):**

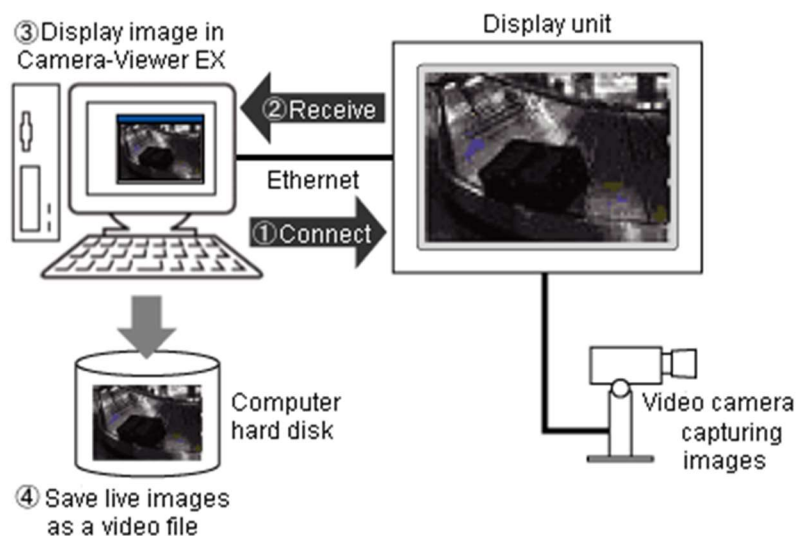
OMR (Optical Mark Recognition) technology is a specialized form of optical scanning utilized for identifying marks made by pens or pencils. It is particularly useful in scenarios where respondents need to select one option from a set of alternatives. OMR systems are commonly employed in educational settings to process examination answer sheets containing multiple-choice questions efficiently.

### **OUTPUT DEVICES:**

Output devices receive binary data from the computer and translate it into a format understandable by humans. They then present this converted information to the external world. Common output devices include video display units, printers, and plotters, each serving distinct purposes in conveying digital data into readable or tangible formats for users.

### **Video display unit:**

A video display unit, reminiscent of a television screen, serves as a visual interface connected to a computer system. This device, also known as a raster scan device, renders digital information into visible images, providing users with a graphical representation of data and facilitating interaction with the computer interface.



## Printers:

There are several types of printers available, each designed for specific printing needs:

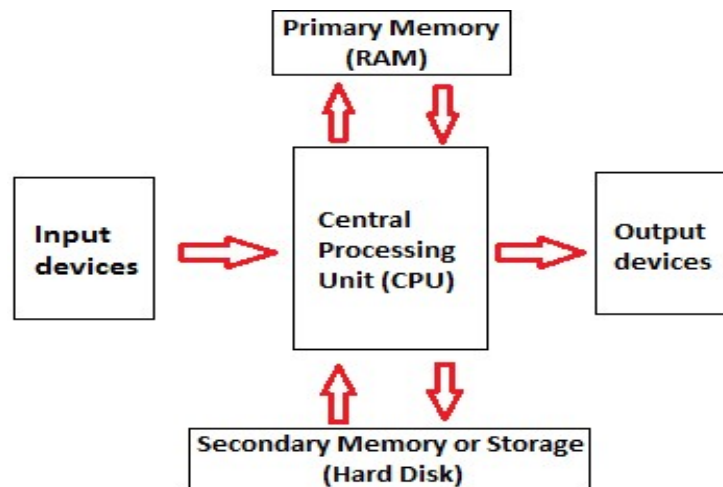
**Character Printer:** These printers operate by printing one character at a time, making them suitable for applications requiring limited text output.

**Line Printer:** Line printers are capable of printing one line of text at a time, offering faster printing speeds compared to character printers and suitable for moderate-volume printing tasks.

**Page Printer:** Page printers are designed to print an entire page at once, providing efficient and high-quality output ideal for documents, graphics, and images.



## COMPONENTS OF CPU AND ITS FUNCTIONS



Every computer has three major components. they are:

- CPU.
- INPUT UNIT.
- OUTPUT UNIT.

## CENTRAL PROCESSING UNIT (CPU):

The Central Processing Unit (CPU), often referred to as the computer's brain, plays a pivotal role in controlling program execution, performing calculations, and overseeing various computer operations. It encompasses several key components, each serving distinct functions:

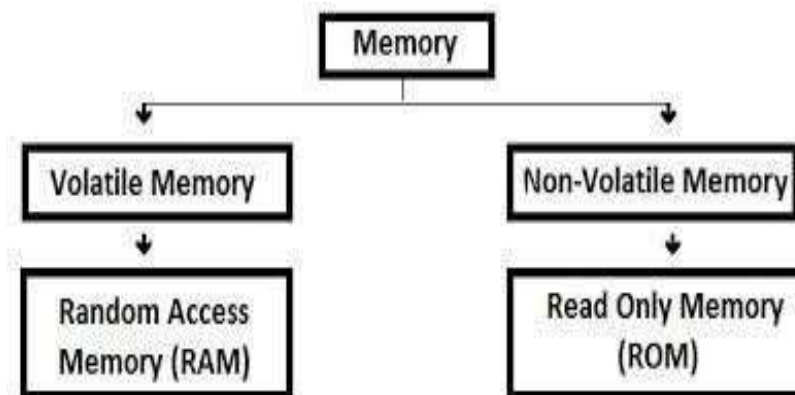
CPU Consist of following parts and Its Functions:

- a) Memory Unit.
- b) Arithmetic and Logic Unit
- c) Control Unit

### a) Memory Unit:

Memory serves as a fundamental component of computer systems, playing a crucial role in storing and retrieving data. It significantly influences the performance of the computer. Memory can be categorized into two main types:

- **Primary Memory/Volatile Memory:** Also known as primary storage, this type of memory is volatile, meaning it loses its stored data when power is turned off. It includes Random Access Memory (RAM), which is used by the computer to temporarily store data and program instructions during operation.
- **Secondary Memory/Non-Volatile Memory:** Secondary storage, or secondary memory, is non-volatile, meaning it retains stored data even when power is turned off. Examples include hard disk drives (HDDs), solid-state drives (SSDs), optical discs (such as CDs and DVDs), and USB flash drives. Secondary memory is used for long-term storage of data and programs.



## **b) Arithmetic and Logic Unit**

The Arithmetic and Logic Unit (ALU) serves as the computational powerhouse within the CPU, responsible for executing a wide range of arithmetic and logical operations. These operations include:

Arithmetic Operations:

- Addition
- Subtraction
- Multiplication
- Division

Logical Operations:

- Comparison between numbers, letters, and special characters
- Logical AND
- Logical OR

Additionally, the ALU handles various conditional operations, including:

- Equal-to condition
- Less-than condition
- Greater-than condition

Overall, the ALU plays a critical role in processing data and executing complex calculations necessary for the functioning of the computer system.

## **c) Control Unit**

The Control Unit serves as the central coordinator within the computer system, directing and orchestrating the activities of other units. Its responsibilities include:

- Instructing the input unit on where to store data received from users.
- Regulating the flow of data and instructions between the storage unit and the Arithmetic and Logic Unit (ALU).
- Overseeing the transmission of results from the ALU back to the storage unit.

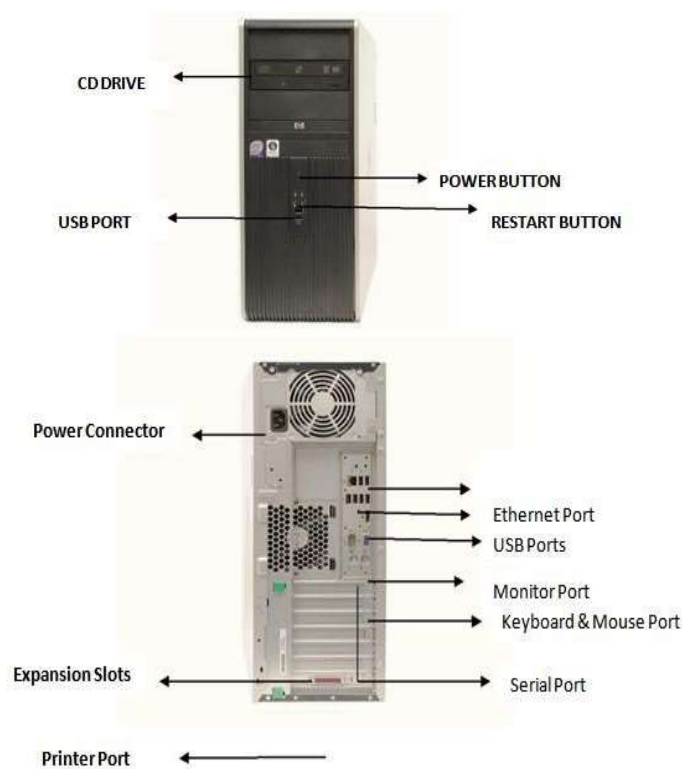
Often likened to the central nervous system of the computer, the Control Unit ensures the smooth and synchronized operation of the entire system, playing a vital role in managing the flow of information and facilitating efficient processing.

## DISASSEMBLING AND ASSEMBLING THE PC

Before delving into the disassembling and assembling process, it's essential to understand several key components of a computer system: Cabinet, Mother Board, Processor, SMPS, RAM, HDD.

### CABINET:

The cabinet serves as the protective outer shell of a computer system, safeguarding vital components such as the motherboard, processor, and others from environmental factors like dust, heat, and moisture. It acts as a shield, preserving the integrity and longevity of the internal hardware components, ensuring optimal performance and reliability.

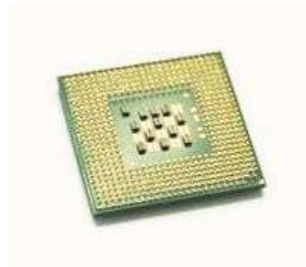


### MOTHERBOARD



The motherboard serves as the central hub of a computer system, housing essential components such as the CPU, memory modules, connectors for storage drives, expansion slots for additional hardware, and ports for external devices like USB peripherals. It acts as a vital interface, directly or indirectly connecting every part of the computer, facilitating seamless communication and coordination between hardware components for efficient operation.

### **CPU/PROCESSOR**



The central processing unit (CPU), or processor, resides within the computer case, typically situated on the motherboard. It appears as a compact ceramic square, housing a silicon chip approximately the size of a thumbnail. Inserted into the motherboard's CPU socket, the CPU operates alongside a heat sink, responsible for dissipating heat generated during operation.

The speed of a processor is gauged in megahertz (MHz), representing millions of instructions per second, or gigahertz (GHz), indicating billions of instructions per second. A higher processor speed denotes the capability to execute instructions more rapidly, enhancing overall system performance.

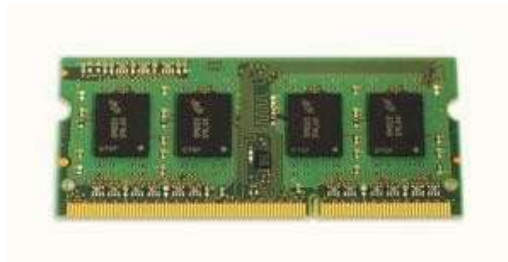
Various manufacturers produce processors for personal computers, with Intel and AMD being the most prominent names in the industry. These companies specialize in designing and manufacturing CPUs tailored to meet diverse computing needs and preferences.

### **POWER SUPPLY UNIT**



The power supply unit (PSU) in a computer is responsible for transforming the electrical power from the wall outlet into the specific type of power required by the computer's components. It distributes this power through cables to various parts of the computer, including the motherboard and other internal components, ensuring they receive the necessary electrical supply to function properly.

## RAM (RANDOM ACCESS MEMORY)



RAM, or Random Access Memory, acts as the short-term memory of your computer system. When your computer executes calculations or processes data, it temporarily stores this information in the RAM until it's required for ongoing tasks or operations.

## HARD DRIVE



The hard drive in your computer serves as the primary storage location for software installation, as well as storing your documents and other files. It functions as long-term storage, retaining data even when the computer is powered off or disconnected. When you initiate a program or access a file, the computer retrieves relevant data from the hard drive and temporarily stores it in the RAM for quick access during ongoing tasks. Similarly, when you save a file, the data is copied back to the hard drive for permanent storage. The speed of the hard drive directly impacts the computer's performance, influencing factors such as startup time and program loading speed.

### STEPS FOR DISASSEMBLING:

**Step 1: Unplugging:** Disconnect all cables from the computer, including power, USB, mouse, keyboard, internet, and modem.

**Step 2: Removing Outer Shell/Casing:** Unscrew and remove the side panels of the computer case by sliding them off.

**Step 3: Removing Top Panel and Front Panel:** Slide off the top panel of the computer case. Remove the front panel by pushing in tabs and sliding it forward.

**Step 4: Removing System Fan:** Unplug the system fan from the motherboard. Unscrew and remove the

system fan from the back of the computer.

**Step 5: Removing CPU Fan:** Unplug the CPU fan from the motherboard. Unscrew and remove the CPU fan from the top of the CPU heat sink.

**Step 6: Removing Power Supply (SMPS):** Unplug all wires connected to the power supply. Unscrew and remove the power supply from the back of the computer.

**Step 7: Removing CD/DVD Drive(s):** Unplug the ribbon cable from the back of the drive. Slide out the CD/DVD drive from the inside of the computer.

**Step 8: Removing Hard Disk Drive:** Disconnect cables from the hard drive. Unscrew and remove the hard drive from its slot.

**Step 9: Removing Expansion Cards:** Remove screws from occupied expansion card slots. Carefully pull upward to remove expansion cards.

**Step 10: Removing Connectivity Centre Cables (front and back panel):** Unplug all cables from the connectivity center. Unplug wires from power button, HDD light, and power light.

**Step 11: Removing RAM (Random Access Memory):** Push down on tabs at both ends of RAM to release it.

**Step 12: Removing Power Button; Power LED + HDD LED:** Cut zip tie holding wires for front connectivity center. Press tabs on chassis' side to remove it. Push LEDs from front and button from back to remove.

**Step 13: Removing Motherboard:** Remove screws holding motherboard to frame. Lift motherboard out of the frame.

**Step 14:** Done!



**Case:**

- Metal Right Panel
- Metal Left Panel
- Plastic Front Panel
- Plastic Top Panel
- Metal & Plastic Frame

**Hardware:**

- Hard Drive
- CD/DVD Drive
- Power Supply
- Expansion Cards
- RAM Chips
- Connectivity Center Cables/Wires:
  - SATA Cable (Hard Drive to Motherboard)
  - Portable HDD Dock and Wires (Power and Data from Portable HDD to Motherboard)
  - Drive Ribbon (CD/DVD Drive to Motherboard)

**Miscellaneous:**

- 33 Screws
- Drive Slot Cover
- 2 Expansion Card Slot Covers
- Plastic Piece (referred to as "chassis" for a button and LEDs)
- Large System Fan
- Small CPU Fan
- Portable HDD Slot

**STEPS FOR ASSEMBLING**

- a. Get Ready: Find a clean, well-lit area and gather all your tools and computer parts.
- b. Put in the Power Supply: Slide the power supply into the case and screw it in place.
- c. Attach the Motherboard: Put the motherboard inside the case and screw it onto the raised spots.
- d. Install the Processor (CPU): Open the CPU socket, put the CPU in correctly, and close the socket.
- e. Attach the CPU Cooler: Apply thermal paste, attach the cooler to the CPU, and secure it.

- f. Add Memory (RAM): Open the slots, put the RAM sticks in carefully, and close the slots.
- g. Install Storage Drives: Slide the hard drive or SSD into its slot and connect the cables.
- h. Connect Front Panel Wires: Attach the case's power button, lights, and other wires to the motherboard.
- i. Put in Expansion Cards: Insert any extra cards like graphics cards and secure them.
- j. Connect Case Fans: Put in any fans and connect them to the motherboard.
- k. Organize Cables: Keep the cables tidy by bundling them together with ties.
- l. Check Everything: Double-check all connections and make sure everything is in place.
- m. Close the Case: Put the side panels back on and screw them shut.
- n. Turn It On: Plug in the computer, press the power button, and make sure it starts up.
- o. Install Software: Insert the operating system disk and follow the instructions to install it.
- p. Install Drivers: Install any extra software needed for your hardware to work properly.

## **WINDOWS INSTALLATION**

Windows 10, part of the Microsoft Windows operating system family, is the latest version designed for desktop computers. It was released to the public on July 15, 2015. There are two main editions: Windows 10 Home, for personal use, and Windows 10 Professional, offering extra features like dual-processor support and domain joining for centrally managed networks.

### **Installation Process:**

Installing Windows 10 typically takes about 1 hour and 30 minutes. To simplify the process, it's broken down into steps. Here's how to begin:

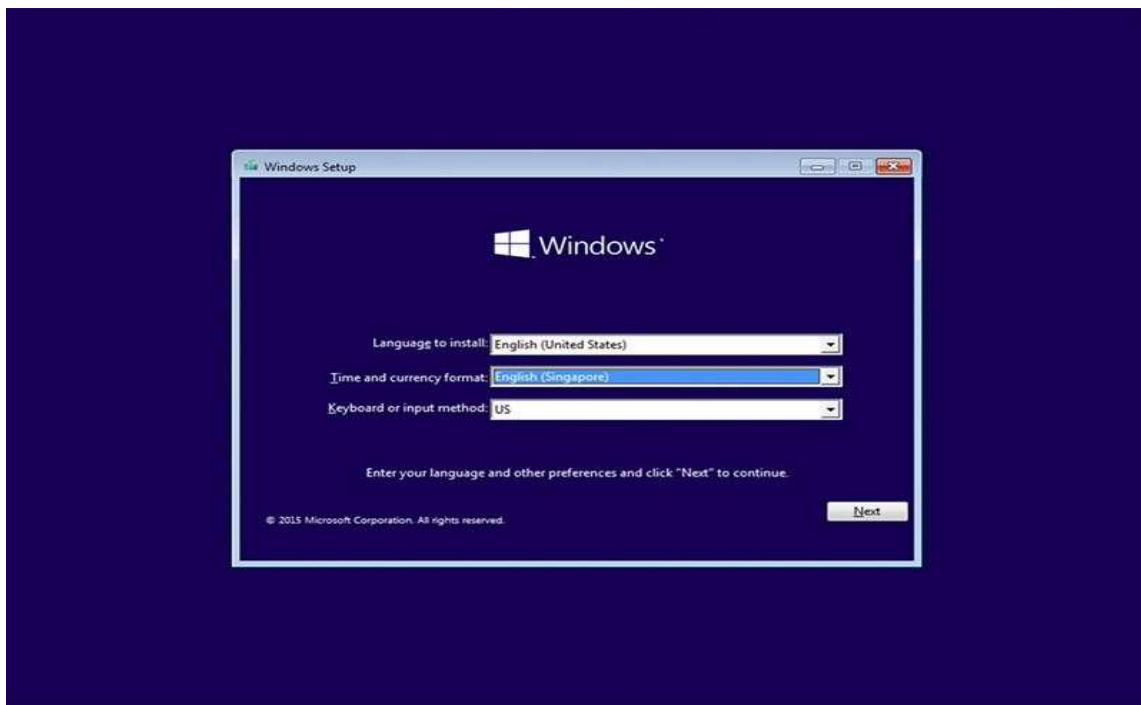
Step 1: Insert the Windows 10 CD or USB drive into your computer and restart it to begin the installation process.



After inserting the Windows 10 CD or USB drive and restarting your computer, follow these steps:

Step 2:

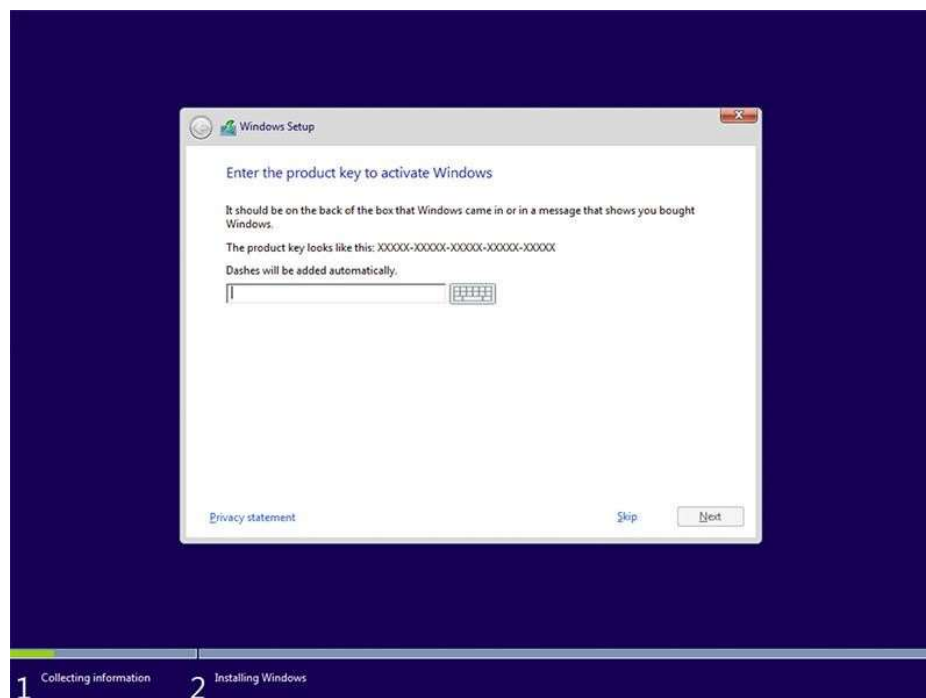
- Upon restart, press and hold the "Del" or "F2" key to enter setup.
- You'll be greeted with the initial setup screen if you're installing Windows 10 from a bootable USB flash drive or DVD.
- Here, you can select the language, time and currency format, and input method for the operating system.



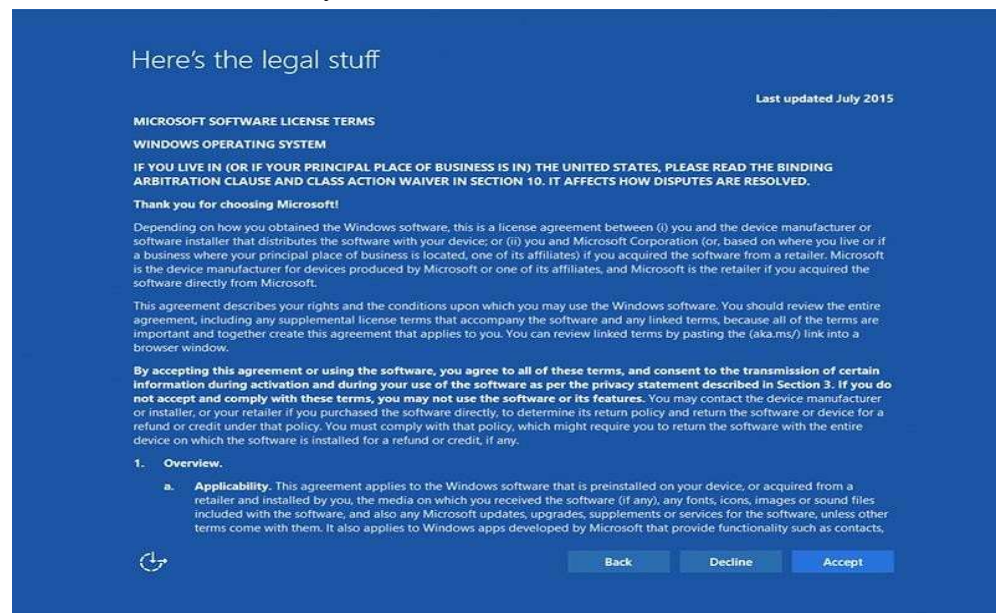
Step 3: You will get the below screen



Step 4: We have to Enter the Product Key



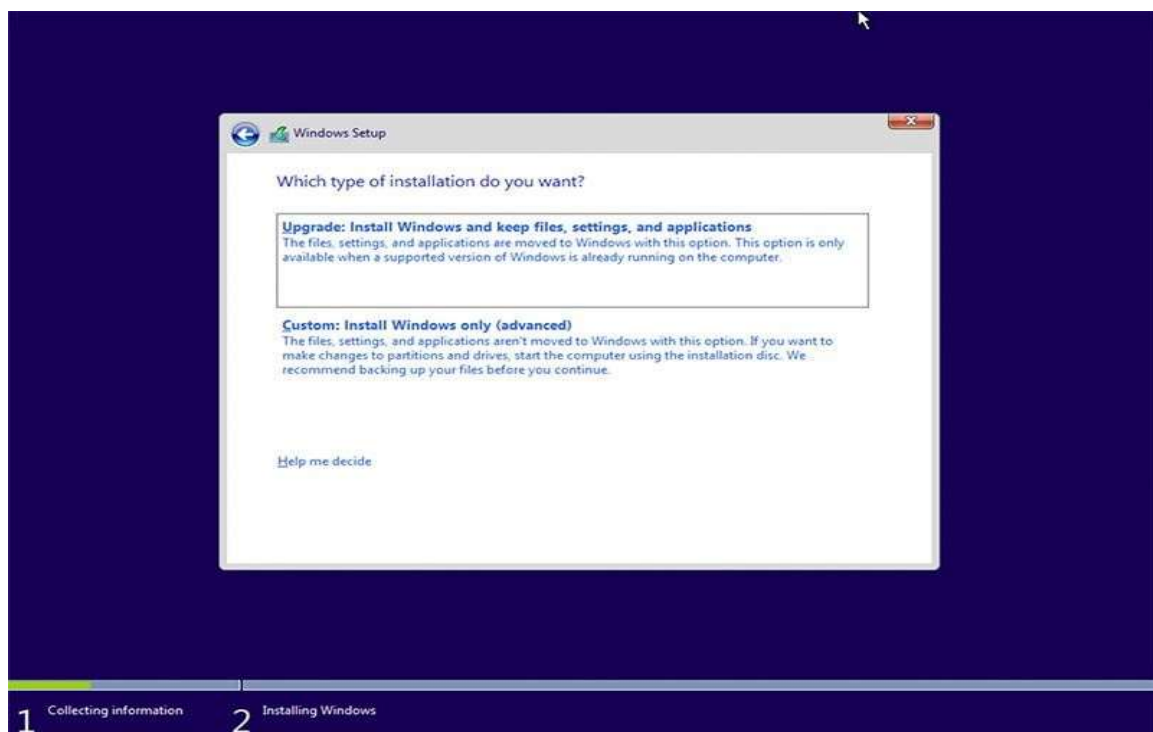
Step 5: Accept the license terms. After you read them.



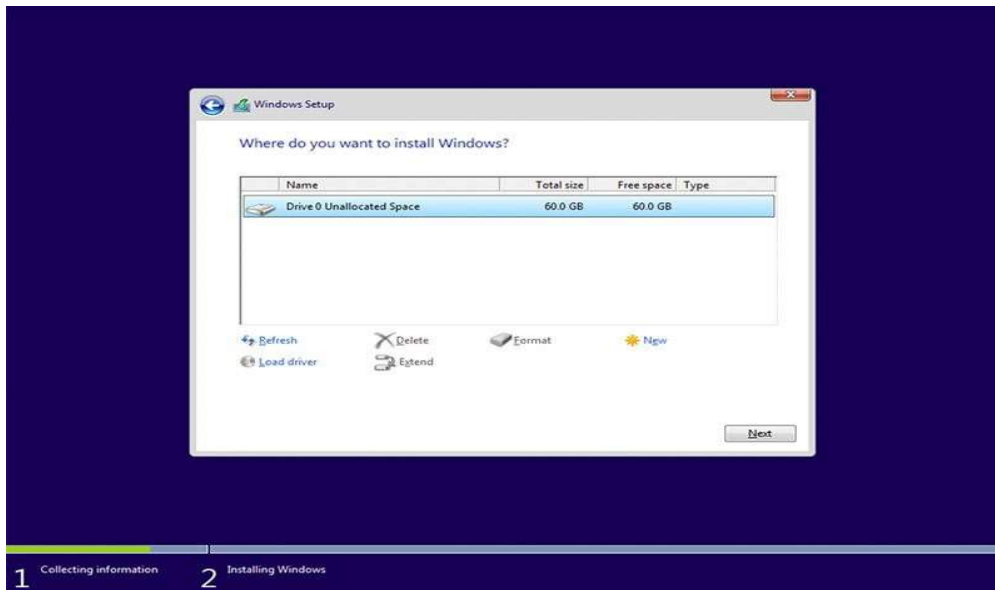
Step 6: Following the initial setup screen, you'll encounter a screen where you have the option to choose between an upgrade or a custom installation.

Opting for an upgrade will transfer files, settings, and apps to Windows.

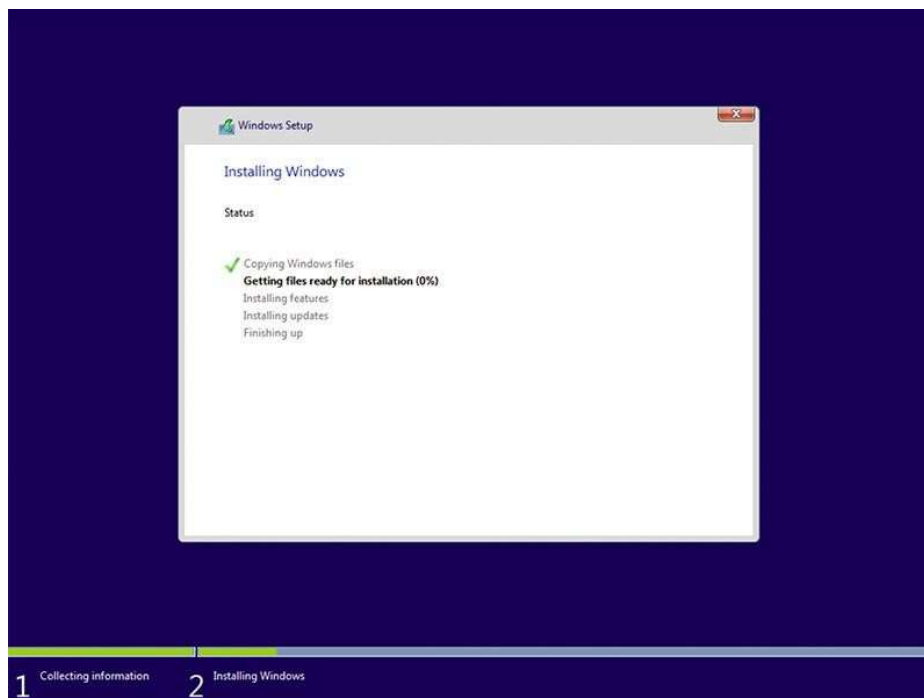
Opting for a custom install means files, settings, and apps won't be moved. This is ideal for a clean install, which is what we're doing.



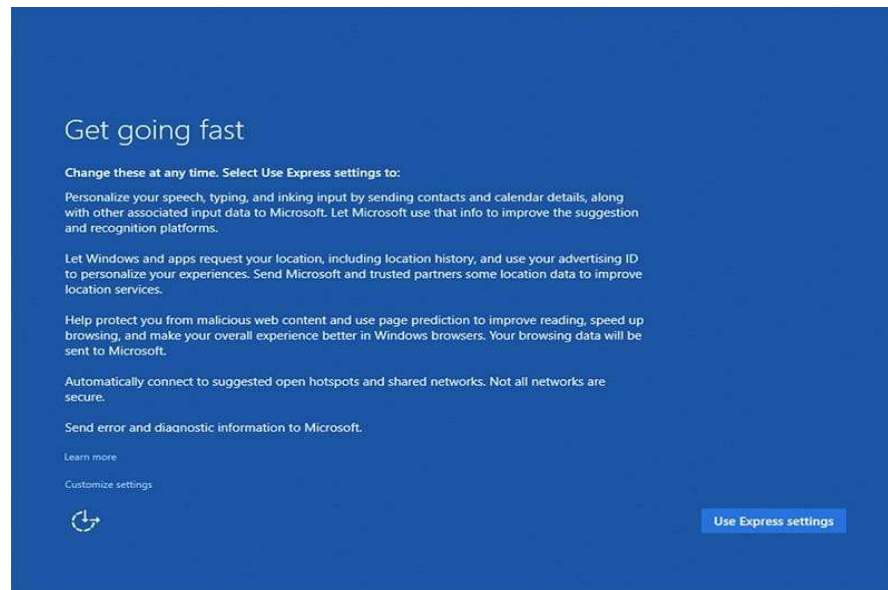
Step 7: This is the screen where you choose the drive to install Windows 10 on. You can also format a drive and manage partitions of the hard drive from here.



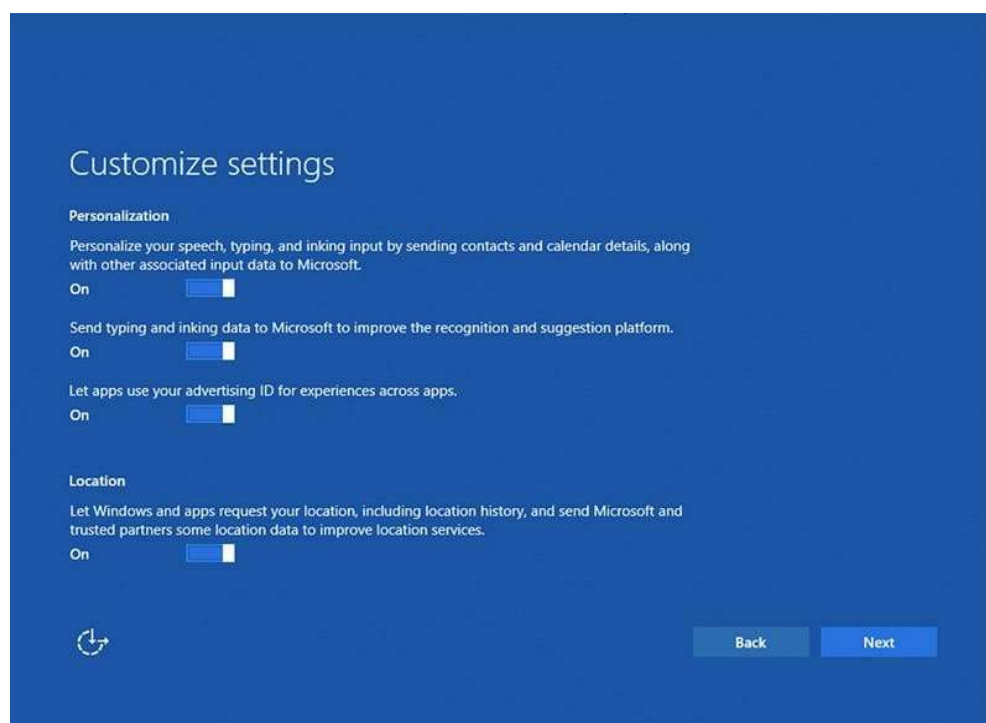
Step 8: Now, you wait.



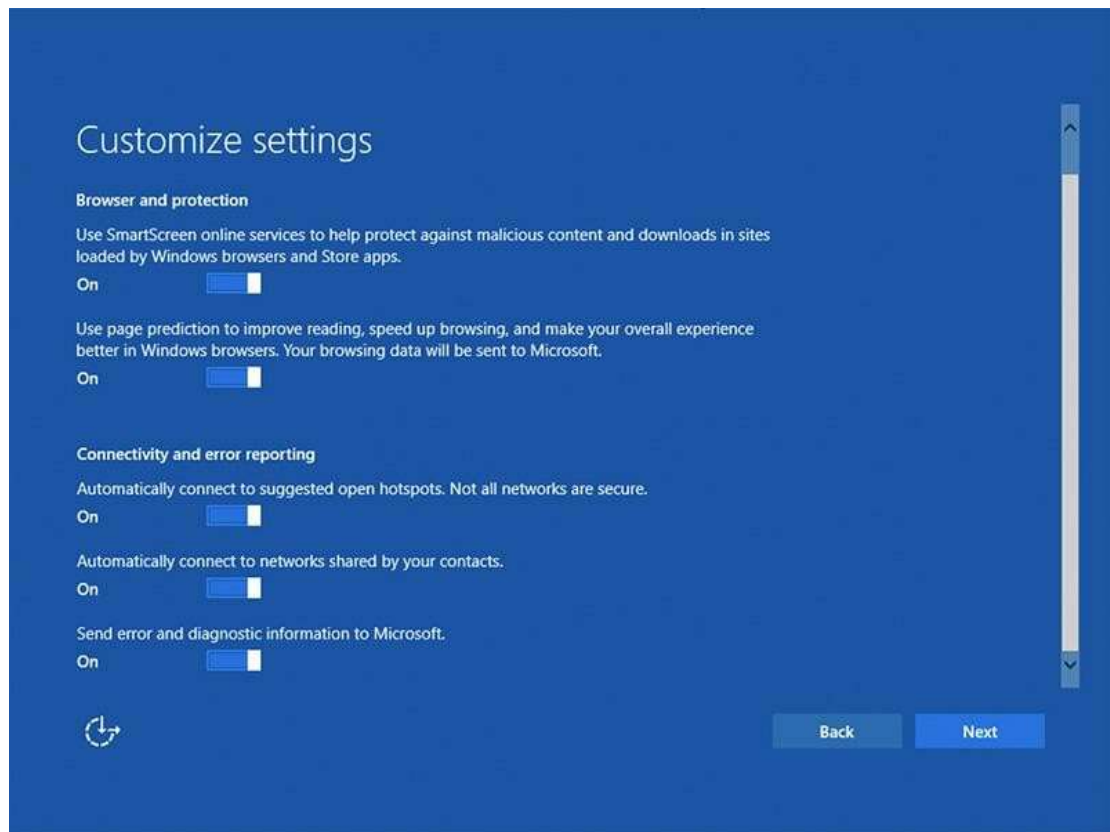
Step 9: Select "Customize settings" to personalize them. Additionally, take note of the small icon located at the bottom left corner. Click on it to access an accessibility menu where you can enable features like increased screen contrast.



Step 10: If you opt to customize the settings, the initial page addresses your contact, calendar, input, and location data. Take your time to review these settings carefully and decide whether you want to enable or disable them according to your preferences.



Step 11: The next page deals with browser data, connectivity, and error reporting.



## USER ACCOUNT CREATION & PERMISSIONS

Admin

Multiple User

## Ubuntu (LINUX) Installation

Ubuntu is a Linux operating system and distribution designed for personal computers, smartphones, and network servers. It utilizes the Unity desktop environment by default and is built upon Debian-based architecture. Ubuntu is renowned for its commitment to free software and is named after the Southern African philosophy of ubuntu, emphasizing the interconnectedness and shared humanity among individuals.

Installation process:

1. Boot Computer with Ubuntu 12.10 Installation CD/DVD or ISO.

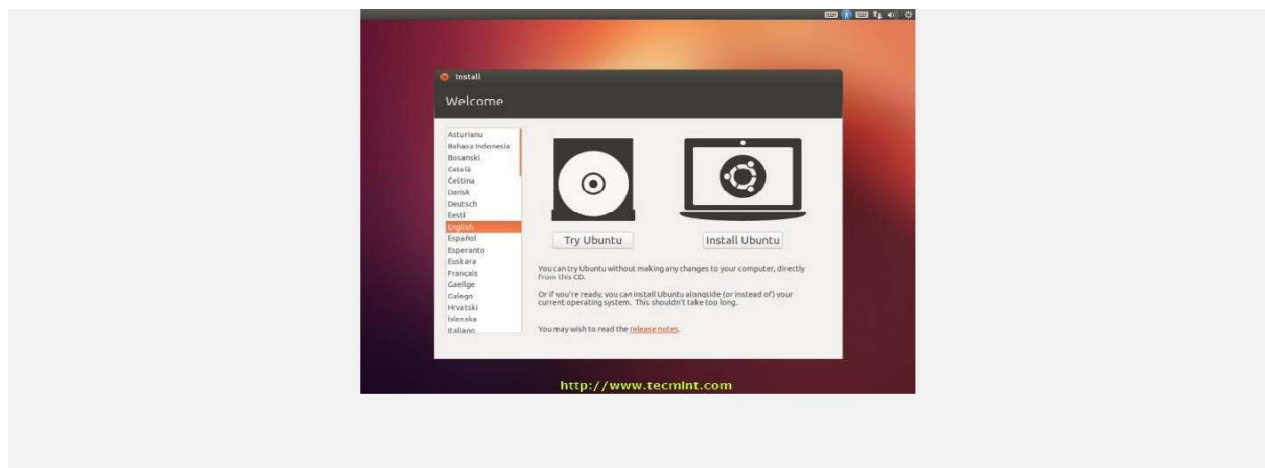


*Boot Ubuntu 12.10*

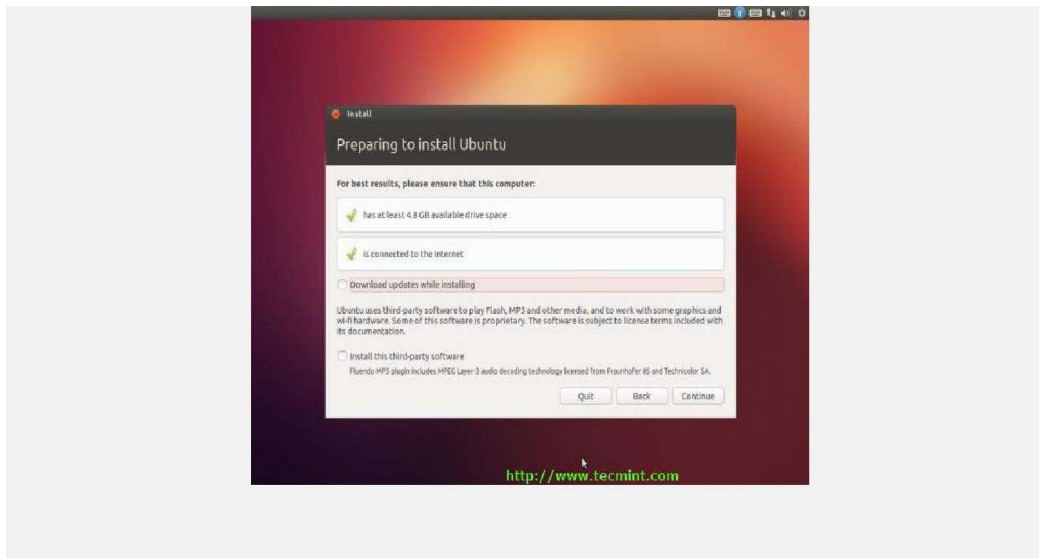
2. Welcome screen of Ubuntu 12.10

*Ubuntu 12.10 Welcome Scree*

Preparing to install Ubuntu, select options and click continue.

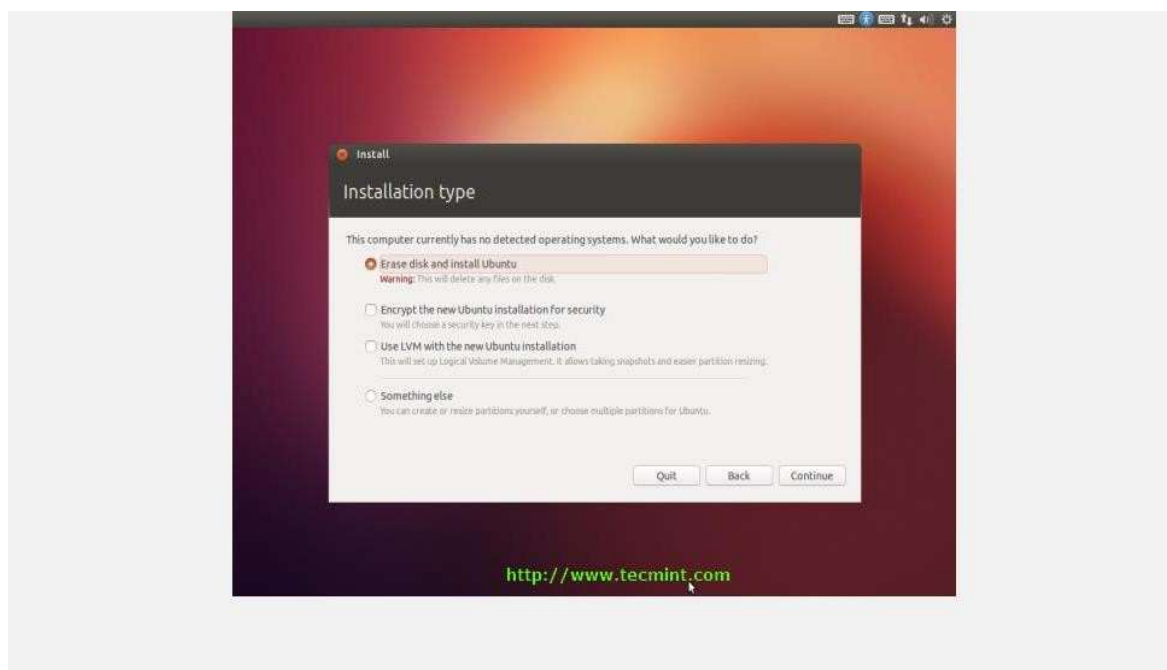


*Preparing Ubuntu 12.10 Installation*



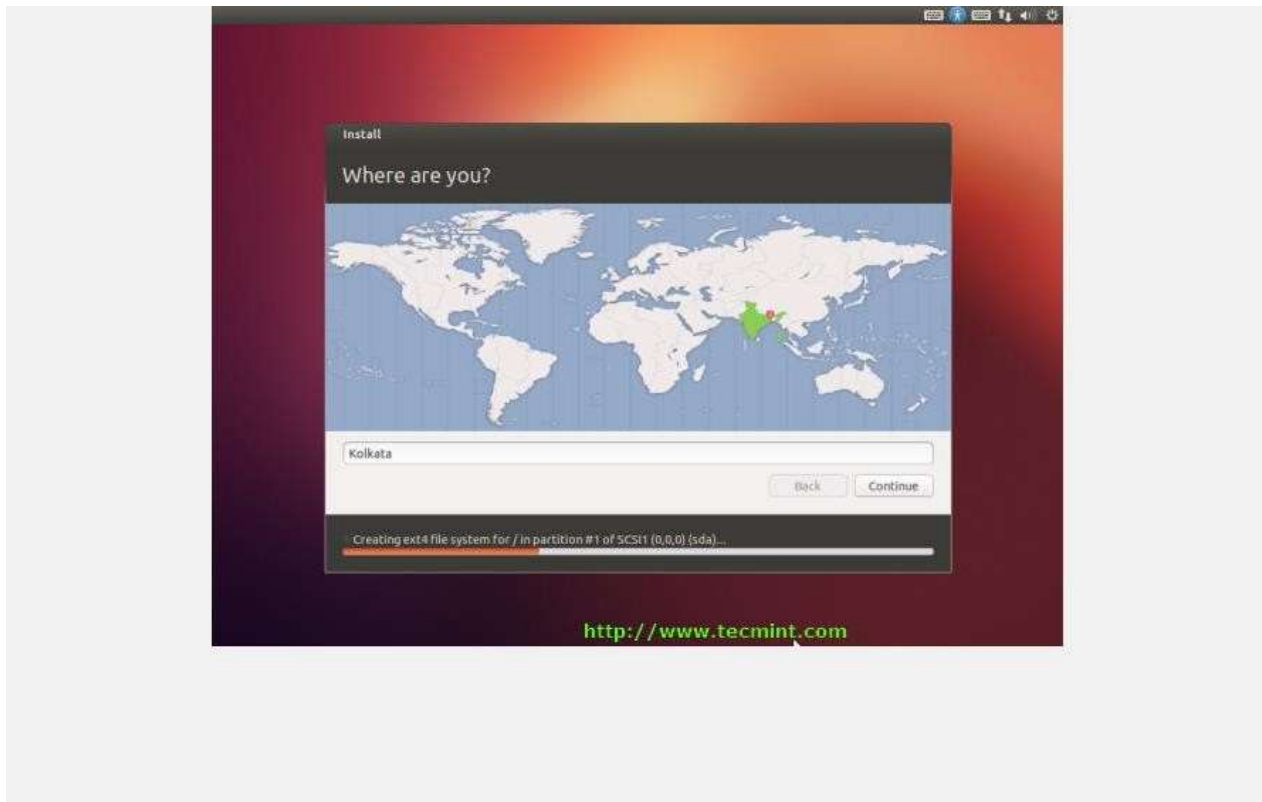
### *Preparing Ubuntu 12.10 Installation*

#### 3. Installation type selection.



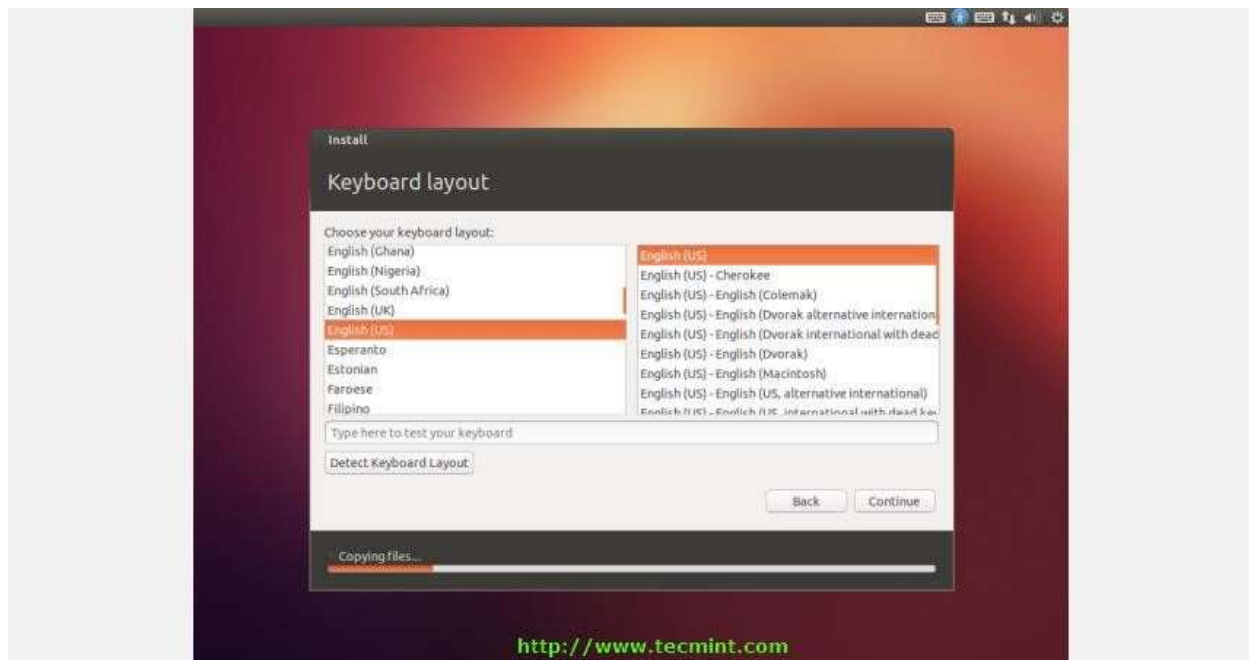
### *Select Ubuntu 12.10 Installation Type*

4. Select nearest city in your time zone.

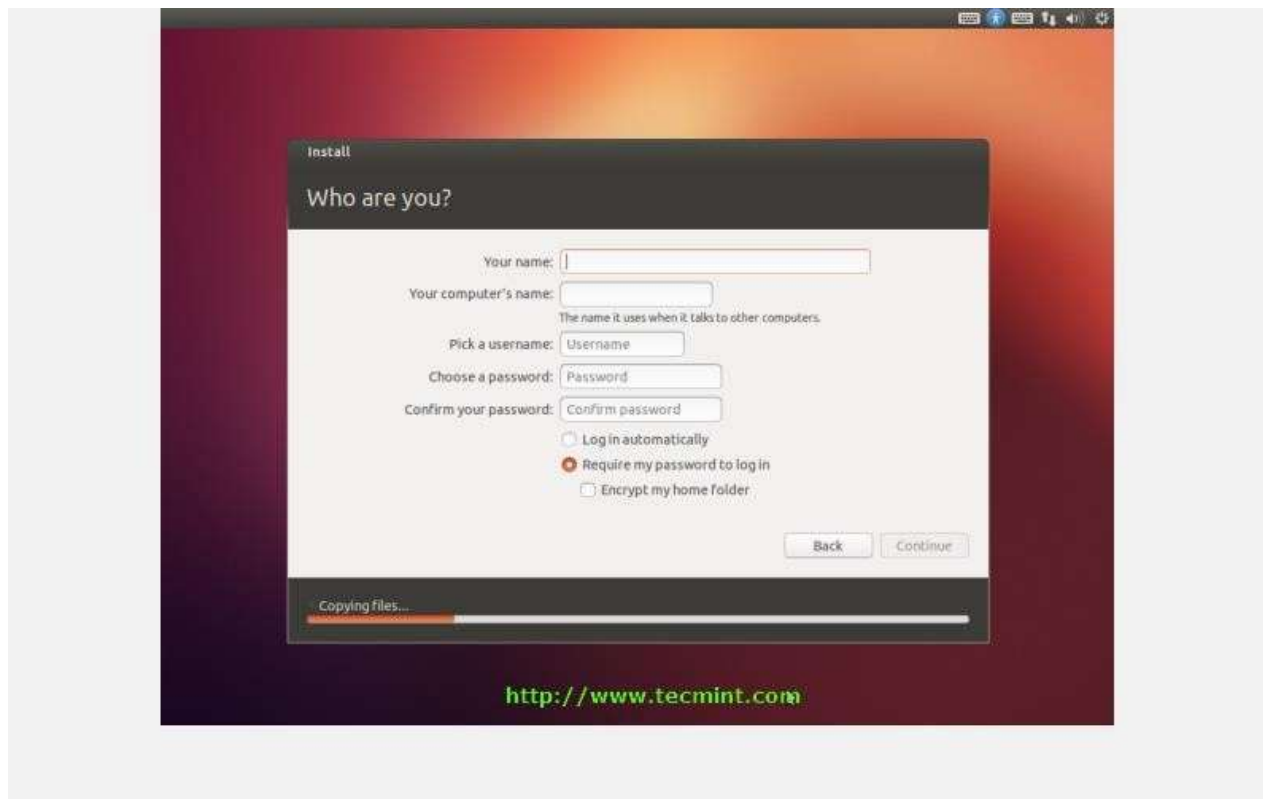


*Select Ubuntu 12.10 Time Zone*

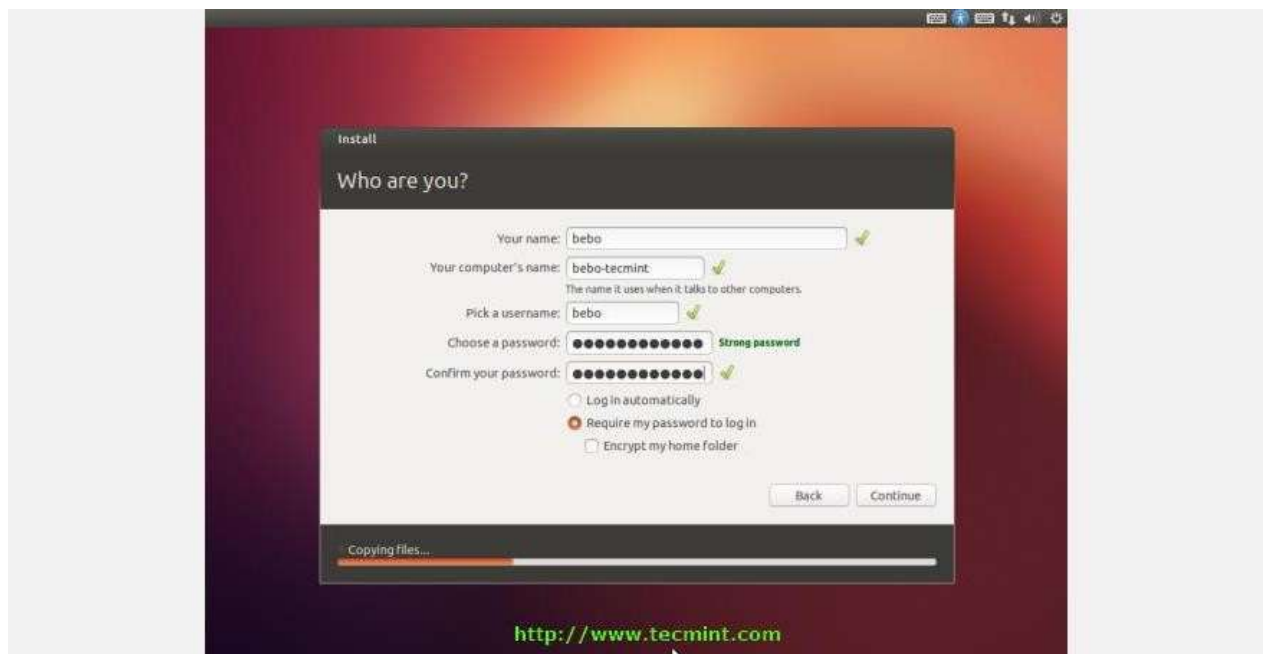
Keyboard selection.



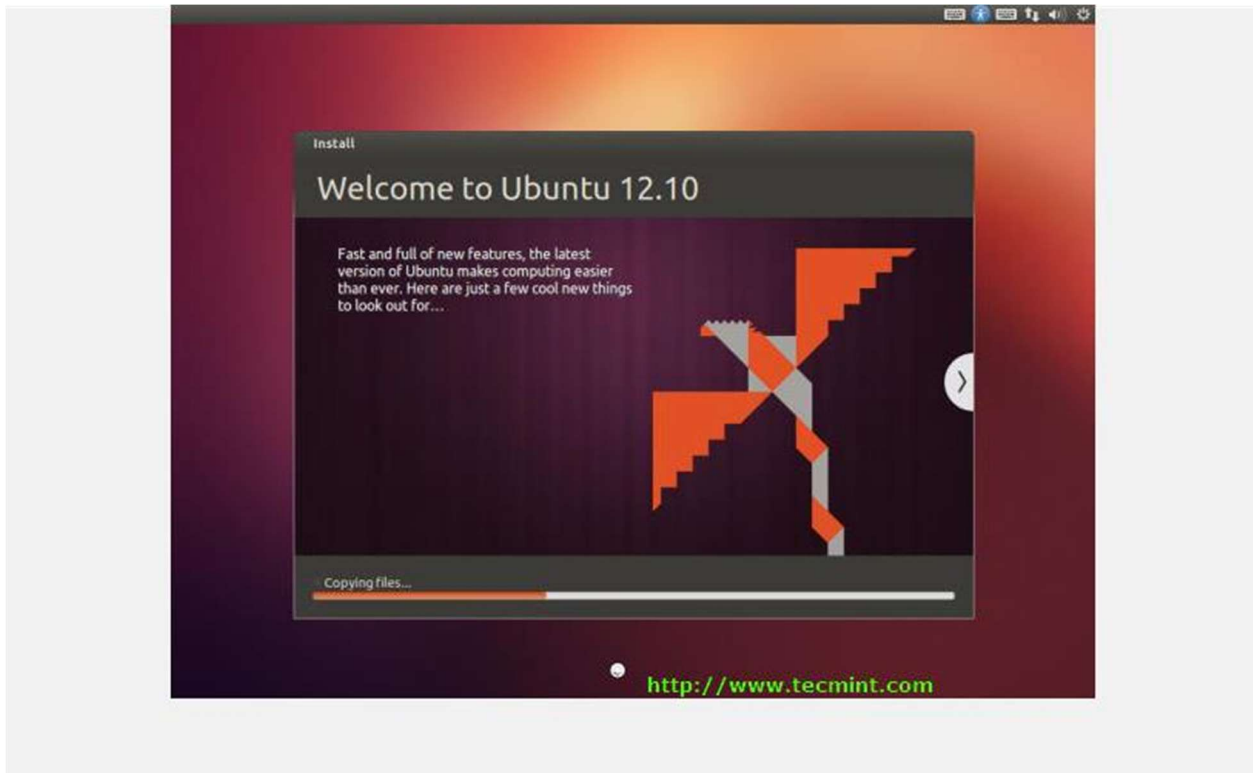
6. Fill information and continue.



7. Installation is in progress. This may take several minutes.



9. Installation is completed, eject CD/DVD and restart.

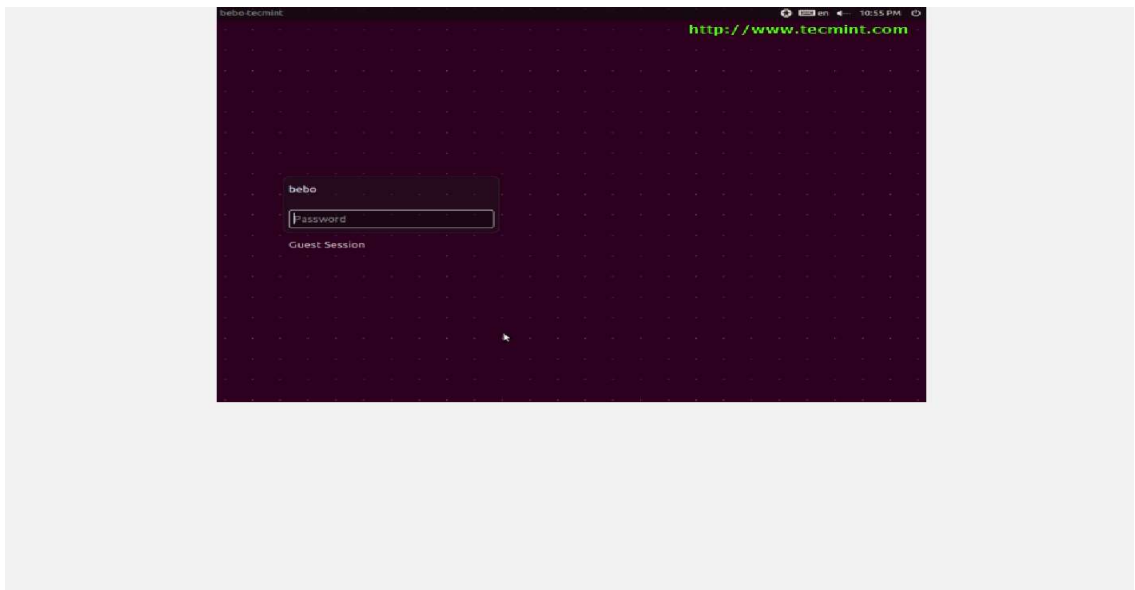


*Ubuntu 12.10 Installation Completed*

10. Booting your Ubuntu Box.



Login screen.

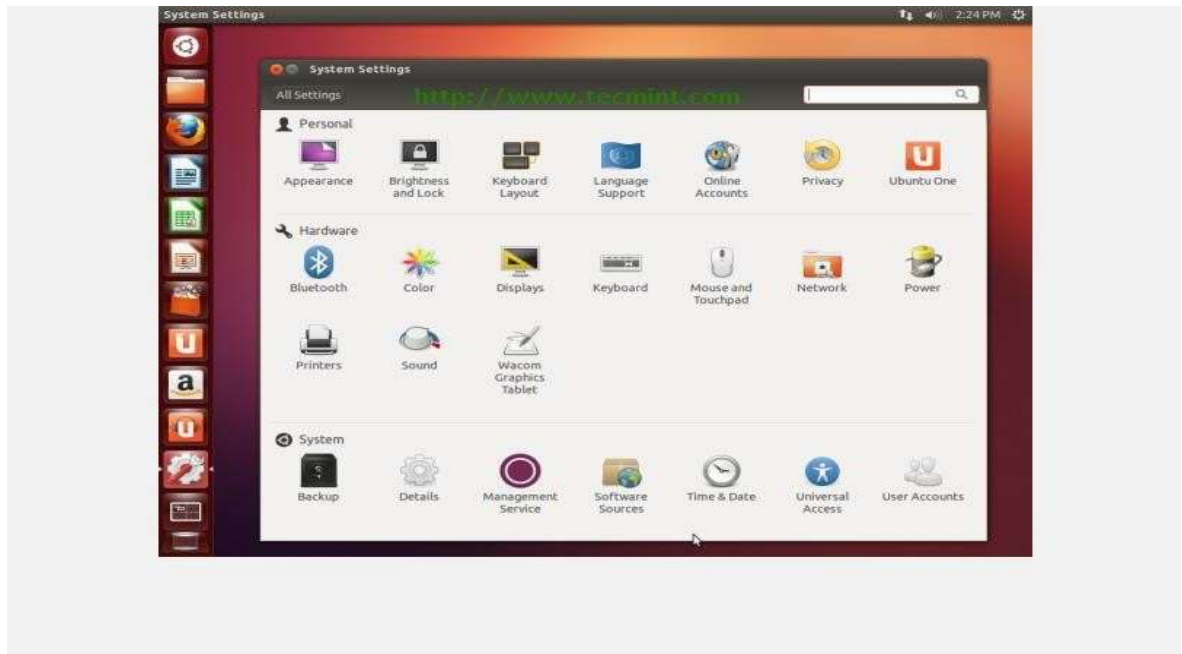


*Ubuntu 12.10 Login Screen*

*12.Ubuntu Desktop.*

### *Ubuntu 12.10 Desktop*

13. Ubuntu Desktop with system settings.



### *Ubuntu 12.10 System Setting*

That's it. Ubuntu is now installed.